

UNIT-01

1-Write and explain the Linux commands to: (a) Display the present working directory (b) Create, view, and navigate directories (c) Display system information
2-Compare and contrast multitasking and multiprogramming, considering their respective strengths and weaknesses.
3-Explain the concept of a batch operating system. Discuss its advantages and limitations in modern computing environments.
4-Describe the architecture of a multiprogramming operating system. How does it improve CPU utilization compared to a single-program system?
5-Write a shell script that uses a function to calculate the factorial of a number entered by the user, but only if the number is prime. If the entered number is not prime, display an appropriate message.
6-Discuss the role of a kernel in an operating system. Explain the differences between a monolithic kernel and a microkernel.
7-Discuss real-time operating systems (RTOS). Provide examples of applications where RTOS is essential.
8-Describe the process of making a system call in an operating system. Include the steps involved and the role of the kernel in handling the call with example.
9-Explain the need of various types of system call in an operating system.
10-Discuss the importance of shell scripting in Linux. Provide an example of a script that uses variables, loops, and conditional statements.
11-How does the distinction between kernel mode and user mode function as a rudimentary form of protection (security) system.
12-Explain how the fork() and exec() system calls are used together in Linux to create and execute new processes. Then, write a simple C program that: - Creates a child process using fork(). - The child process replaces itself using exec() to run the ls command. - The parent process waits for the child to finish before exiting. Now consider the following code snippet using the fork() and wait() system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors. <pre>int x = 4; while(x > 0) { fork(); printf("hello"); wait(NULL); x--; }</pre> The total number of times the printf statement is executed is ____
13-Describe the architecture of a multiprocessor operating system. How does it enhance system performance compared to a single-processor system?

14-Complete the following shell script to find the average of numbers given at the command line.

```
#!/bin/bash
sum=0
count=$#

for num in "$@"
do
    sum=$(( sum + num ))
done

avg=$(( sum / count ))

echo "Average = $avg"
```

15-Explain the concept of a real-time operating system (RTOS). Discuss the differences between hard real-time and soft real-time systems with examples.

16-Describe the working of kernel in an operating system. Explain how it manages communication between hardware and software components.

17-Describe the role of the Command Line Interface (CLI) in Linux compared to the Graphical User Interface (GUI). Also summarize the purpose of basic Linux commands such as ls, ls-l, mkdir, rmdir, cp, and rm in Linux.

18-Write a shell script to calculate the Fibonacci sequence. Explain each step of the script.

19-Predict the output of the following shell script when executed as:

```
#!/bin/bash
num=$1
pos=1

while [ $num -ne 0 ]
do
    digit=$((num % 10))
    if [ $((pos % 2)) -ne 0 ]
    then
        echo -n "$digit "
    fi
    num=$((num / 10))
    pos=$((pos + 1))
done
echo
```

20-Write a shell script to display the digits which are in odd position in a given 5-digit number. Explain each step of the script.

21-Give the type of OS that will be used in each case and give reason. 1)- Air Traffic Control Systems 2)- Payroll system 3)- Medical imaging systems 4)- Server 5)- Robots 6)- Bank Statements 7)- Cloud 8)- Weapon and missile systems

UNIT-02

1-Use the Round Robin scheduling algorithm to schedule five processes and compute the turnaround time.

2-Difference between Multilevel Queue and Multilevel Feedback Queue Scheduling.

3-Explain Round Robin Scheduling with an example.

4-Describe the various states in a Process Control Block (PCB).

5-Explain the concept of starvation in scheduling and methods to handle it.

6-Compare and contrast SJF and Round Robin scheduling with examples.

7-Explain the structure of a Linux process management system.

8-Discuss the different types of process scheduling queues.

9-Consider the following set of processes with their respective arrival times (AT) and burst times (BT), scheduled using the First-Come, First-Served (FCFS) algorithm:
Calculate:

1.The completion time (CT) for each process.

2.The average waiting time (AWT).

3.The average turnaround time (TAT).

Show your calculations.

10-Consider the following set of processes with their respective arrival times (AT) and burst times (BT). The scheduling follows the First-Come, First-Served (FCFS) algorithm. Tasks:

1.Construct a Gantt Chart for this scheduling.

2.Calculate the Completion Time (CT) for each process.

3.Determine the Turnaround Time (TAT) and Waiting Time (WT) for each process.

4.Compute the Average Waiting Time (AWT) and Average Turnaround Time (ATAT).

5.Interpret the results: How does the arrival time of P3 (which is earlier than P4) affect the overall waiting time of processes?

11-Using FCFS, calculate for each process:

♦Completion Time (CT)

♦Turnaround Time ($TAT = CT - AT$)

♦Waiting Time ($WT = TAT - BT$)

Also calculate:

♦ATAT (Average Turnaround Time)

♦AWT (Average Waiting Time)

ACT (Average Completion Time)

12-Use SJF (non-preemptive) to calculate for each process:

♦Completion Time (CT)

♦Turnaround Time ($TAT = CT - AT$)

♦Waiting Time ($WT = TAT - BT$)

Also calculate:

♦ATAT (Average Turnaround Time)

♦AWT (Average Waiting Time)

♦ACT (Average Completion Time)

13-Time Quantum: 3 milliseconds Draw the Gantt Chart, then compute:

♦Completion Time (CT)

♦Turnaround Time (TAT)

♦Waiting Time (WT)

Averages (ATAT, AWT, ACT)

14-Use Preemptive SJF (SRTF) to calculate CT, TAT, WT, ATAT, AWT, ACT.

15-Note: Lower number = higher priority

Task:

Schedule the processes using non-preemptive priority scheduling. Calculate CT, TAT, WT, ATAT, AWT, ACT.

16-Time Quantum: 2 milliseconds

Task:

Draw the Gantt Chart and compute CT, TAT, WT, ATAT, AWT, ACT.

17-Consider the set of 5 processes whose arrival time and burst time are given below- If the CPU scheduling policy is priority non-preemptive, calculate the average waiting time and average turnaround time. (Higher number represents higher priority)

18-Solve for the average waiting time and turnaround time using the First Come First Serve (FCFS) algorithm for the following processes:

19-Compute the average turnaround time using the Shortest Job First (SJF) non-preemptive algorithm for these processes:

20-Analyze the response time and waiting time using Shortest Remaining Time First (SRTF) for the given dataset:

21-Derive the execution order and CPU utilization using preemptive priority scheduling where lower numbers indicate higher priority:

22-Describe the Process Transition Diagram with a labeled diagram.

UNIT-03

1-Discuss RAG and give example of wait-for graph (WFG).
2-Explain any two deadlock handling methods in detail.
3-Discuss the classical problems of synchronization in detail.
4-Discuss the atomic operations of semaphore and show how mutual exclusion can be implemented.
5-Discuss the process of deadlock detection and also explain various approach we follow to avoid deadlock detection in detail.
6-Discuss about Peterson solution for the critical section problem in detail.
7-Describe Lamport Bakery Solution to the critical section problem.
8-Discuss Sleeping Barber problem in detail.
9-Elaborate the reader-writer problem in detail
10-Elaborate Race Condition and how synchronization helps in concurrent programming.
11-Write a short note on Interprocess communication with its advantages and disadvantages
12-1Show how mutual exclusion can be implemented and give example of synchronization hardware using Test and Set operation.
13-Describe the Bounded - buffer problem and give a solution for the same using semaphores
14-Write code for Dining Philosopher problem and discuss the approaches for reducing deadlock condition.
15-Discuss in detail the critical section problem and also write the algorithm for Readers/Writers Problem with semaphores.
16-Mention five real life examples of deadlock situation and also suggest the solution of mentioned problems.
17-Describe Banker's algorithm with an example.

18-Assume a system with four resource types R1, R2, R3 and R4 with these many units (6,4,4,2), respectively, and the maximum claim on resources and the current allocation given below. Is this state safe? Explain.

Maximum request

Process R1 R2 R3 R4

P1 3 2 1 1

P2 1 2 0 2

P3 1 1 2 0

P4 3 2 1 0

P5 2 1 0 1

Current Allocation

Process R1 R2 R3 R4

P1 2 0 1 1

P2 1 1 0 0

P3 1 1 0 0

P4 1 0 1 0

P5 0 1 0 1

19-If we ran processes sequentially, would we ever have deadlocks in the system? Justify your answer

19-In a uniprocessor system, can the system ever enter into a deadlock? Justify your answer.

20-A club has a lounge where the members can sit and chat. Members include both smokers

and non-smokers. Smokers can smoke in the lounge when non-smokers are absent. Devise a protocol for the lounge. A smoker calls `enterLounge(true)` to enter the lounge (the flag `true` indicates that she is a smoker), then calls `smoke()`, and finally calls `leaveLounge(true)` to enter, then sits in the lounge, and finally calls `leaveLounge(false)` to leave the lounge.

Using semaphores, write a code for the functions noted above so that smoking rules are obeyed.

Name and describe three desirable properties that any synchronization algorithm should possess. Explain briefly whether your solution satisfies each property.

21-Consider a traffic crossing between two roads, one in the east-west direction and the other in the north-south direction. Suppose the crossing is modelled as a shared data structure, and cars are modelled as processes that access the crossing in order to pass through it. Assume that cars only travel straight, without turning left or right. Two or more cars that are allowed to simultaneously pass through the crossing only if they are headed in the same or opposite directions (e.g., a north-bound and south-bound car), but east-west traffic and north-south traffic can never access the crossings simultaneously. Devise a synchronization solution for the cars using semaphores. You should write two procedures **`east_west()`** and **`north_south()`** to show how cars travelling in the respective direction should behave.

22-Write a monitor that implements an alarm *clock* that enables a calling program to delay itself for a specified number of time units (*ticks*). You may assume the existence of a real hardware clock that invokes a procedure *tick* in your monitor at regular intervals.

23-Consider a system consisting of processes $P_1, P_2, \dots, P_n$, each of which has a unique priority number. Write a monitor that allocates three identical line printers to these processes, using the priority numbers for deciding the order of allocation.

24-Show that the two-phase locking protocol ensures conflict serializability.

UNIT-04

1-Compare FIFO with other page replacement algorithms (like LRU, LFU, and Optimal). Under what situations would FIFO be more efficient than other algorithms?

2-Given a scenario where a system is experiencing thrashing, explain how the operating system can detect and address this problem. Discuss the strategies for reducing or eliminating thrashing, such as page swapping, working set management, and memory allocation tuning.

3-Explain the concept of non-contiguous memory allocation and how it overcomes the limitations of contiguous allocation. Discuss the techniques involved in non-contiguous memory allocation, focusing on Paging and Segmentation. How do these techniques help in minimizing fragmentation?

4-Define segmented paging and explain how it combines the features of both segmentation and paging to manage memory. What are the benefits of segmented paging over pure segmentation and paging, and in what scenarios is segmented paging particularly useful?

5-Discuss the concept of virtual memory and explain how it allows the system to run processes that may not fit entirely into physical memory. What role does demand paging play in virtual memory management? How does demand paging optimize memory usage, and what factors contribute to its performance?

6-Define thrashing in the context of virtual memory systems. What are the causes of thrashing, and how does it affect system performance? What strategies can be employed to reduce or prevent thrashing in an operating system?

7-Discuss address binding in the context of loading and linking, explaining the different types of address binding (compile-time, load-time, and execution-time). How do these bindings affect the execution of a process?
8-In memory management, allocation strategies play a crucial role in determining how memory is assigned to processes. Compare and contrast the First Fit, Best Fit, and Worst Fit strategies for memory allocation. How do these strategies impact the efficiency of memory usage and process execution?
9-Explain the concept of virtual memory and how it allows processes to exceed physical memory limits. Discuss the concept of demand paging and its impact on system performance. How is page fault handling managed in demand paging, and what are the factors influencing its efficiency?
10-Describe the function of memory management in an operating system. Discuss the role of loading and linking in the memory management process. Also, explain address binding and how it occurs at different stages (compile-time, load-time, and execution-time), with examples of how it impacts process execution.
11-Explain the concept of Belady's Anomaly and discuss how it applies to page replacement algorithms. Which algorithms are susceptible to Belady's Anomaly?
12-Consider the following reference string: 4 , 3, 2, 1, 4, 3, 5, 4, 3, 2, 1, 5, and a frame size of 3. Calculate the page fault rate using FIFO. Explain your solution step-by-step.
13-Given a reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2 and a frame size of 3, calculate the number of page faults using FIFO. Show your calculations and explain the results.
14-Given the following reference string: 3, 1, 2, 3, 4, 2, 3, 0, 3, 1, 3, and a frame size of 3, calculate the number of page faults using the LRU.
15-Consider the following reference string: 4 , 3, 2, 1, 4, 3, 5, 4, 3, 2, 1, 5, and a frame size of 3. Calculate the page fault rate using LRU. Explain your solution step-by-step.
16-Given the following reference string: 1, 2, 3, 4, 2, 5, 3, 4, 2, 6, 3, 8, 4, and a frame size of 3, calculate the number of page faults using the Optimal page replacement algorithm.
17-Given the following reference string: 5, 3, 1, 5, 4, 1, 6, 3, 4, 5, 1, and a frame size of 3, calculate the number of page faults using the Optimal page replacement algorithm.
18-Consider the following reference string: 0, 1, 2, 3, 0, 1, 4, 0, 1, 2, 3, 4, and a frame size of 4. Calculate the page fault rate using LRU. Explain your solution step-by-step.
19-Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using-First Fit Algorithm,Best Fit Algorithm,Worst Fit Algorithm
20-Consider the page references 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4-page frame. Find number of page fault using Optimal Page Replacement Algorithm.

21-Consider the following reference string: 1, 2, 3, 4, 5, 1, 2, 3, and a frame size of 3. Calculate the page fault rate using FIFO, LRU, Optimal, and LFU. Compare the results and explain the differences

UNIT-05

1-Describe the key differences between parallel and distributed systems.

2-How do virtual machines help in resource management?

3-Explain the importance of hypervisors in virtualization.

4-What are virtual machines? Differentiate between Type 1 and Type 2 hypervisors.

5-How can GPUs improve the performance of AI and machine learning tasks?
Develop a GPU-based framework for real-time image processing.

6-Why does C-LOOK perform better than LOOK in some scenarios?

7-Compare the architectures of CPU and GPU in terms of processing efficiency.

8-How can GPUs improve the performance of AI and machine learning tasks?

9-A company wants to implement virtualization for cost efficiency. Suggest a suitable hypervisor and justify your choice.

10-Which is better for handling large-scale computations: a parallel system or a distributed system? Justify your answer.

11-Design a basic distributed system model for a real-world application (e.g., online banking).

12-Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The FCFS scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is?

13-Consider a disk with 200 tracks and the queue has random requests from different processes in the order:55, 58, 39, 18, 90, 160, 150, 38, 184 Initially arm is at 100. Find the Average Seek length using FIFO, SSTF, SCAN and C-SCAN algorithm.

14-Disk requests come to a disk driver for cylinders in the order 10, 22, 20, 2, 40, 6 and 38 at a time when the disk drive is reading from cylinder 20. The seek time is 6 ms/cylinder. The total seek time, if the disk arm scheduling algorithms is first-come-first-served is

15-Consider the disk system with 100 cylinders. The request to access the cylinders occur in the following sequence.4, 37, 10,7,19,73,2,15,6,20.Assuming the head is currently at cylinder 50 what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek ime first algorithm is used.

16-Given: A disk queue with requests at the following track numbers: 98, 183, 37, 122, 14, 124, 65, 67 **Initial head position:** 53 **Disk size:** 200 tracks **Head movement direction:** Right. Solve the following:Compute the total seek time using the FCFS (First-Come, First-Served) algorithm.

17-Solve using the SSTF (Shortest Seek Time First) algorithm.

18-Apply the SCAN (Elevator) algorithm and find the total head movement.

19-Solve using the C-SCAN (Circular SCAN) algorithm.

20-Explain the impact of disk architecture on system performance.

21-How does disk cache improve the performance of storage devices?

22-What is disk scheduling? Explain its need in operating systems.

23-2Describe the First-Come, First-Served (FCFS) disk scheduling algorithm with an example.

24-Explain the Shortest Seek Time First (SSTF) scheduling algorithm with a diagram.